

B.E. Even Semester Midterm Examination, 2024-2025

Subject: Operating System | **Paper Code:** PCC-IT 403 | **Full Marks:** 15

Q1. What is system call? (2)

Answer: A system call provides the programmatic interface between a user-space application and the OS kernel. It allows restricted user programs to request the OS to perform privileged tasks like file I/O or memory allocation safely.

Q2. What does short-term scheduler do? (2)

Answer: The short-term scheduler (CPU scheduler) selects one of the processes from the ready queue (processes loaded in memory) and allocates the CPU to it for execution. It runs very frequently.

Q3. Calculate the waiting time for process C when three processes A, B and C come in sequence with CPU burst time 10, 20 and 30s. FCFS is followed. (2)

Answer: FCFS executes in order. Process A waits 0s. (Executes 0 to 10). Process B waits 10s. (Executes 10 to 30). Process C must wait for A and B. Waiting time = $10 + 20 = 30$ seconds.

Q4. State the forms of memory wastage in fixed and dynamic partitioning schemes. (2)

Answer: * **Fixed Partitioning:** Internal Fragmentation (unused space inside an allocated partition). * **Dynamic Partitioning:** External Fragmentation (free memory is broken into small, non-contiguous blocks that cannot be allocated).

Q5. Calculate effective memory access time: 90% TLB hit, main memory = 20ns, TLB = 2ns. (2)

Answer: Formula: $\text{EMAT} = h \times (c + m) + (1 - h) \times (c + 2m)$
 $\text{EMAT} = 0.90(2 + 20) + 0.10(2 + 2(20)) = 0.90(22) + 0.10(42) = 19.8 + 4.2 = 24$ ns.

Q6. What is demand segmentation? (2)

Answer: Similar to demand paging, only the necessary segments (logical units like functions/arrays) of a process are loaded into memory when they are explicitly requested during execution, saving memory.

Q7. Explain the role of PMTBR and PMTLR. (3)

Answer: * **PMTBR (Page Map Table Base Register):** Points to the base physical address of the page table for the running process in main memory. * **PMTLR (Page Map Table Limit Register):** Holds the total size of the page table to prevent a process from accessing memory outside its logical address space (segmentation fault).